

Yiliang Zhou

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Summary

Starting from background in Biochemistry, I am working on the interdisciplinary field of public health, biomedical, and **Informatics** now. My research interests revolve around Natural Language Process, and LLMs application on Biomedical and EHRs. My current research emphasizes promoting equity in healthcare and precise outcome extraction, with a focus on addressing disparities and enhancing decision-making in clinical environments.

Education

University of California, Irvine <i>Ph.D in Informatics</i> GPA: 4.00/4.00	Irvine, CA <i>September 2024 - June 2029 (Expected)</i>
Weill Cornell Medicine - Cornell University <i>Master of Science in Health Informatics</i> GPA: 3.912/4.00	New York, NY <i>September 2022 - August 2023</i>
University of Washington <i>Bachelor of Arts in Biochemistry; Minor in Data Science</i>	Seattle, WA <i>September 2018 - June 2022</i>

Employment

University of California, Irvine <i>Graduate Researcher</i>	<i>September 2024 - Current</i> Irvine, CA
Weill Cornell Medicine <i>Senior Research Associate</i>	<i>March 2024 - August 2024</i> New York, NY
Weill Cornell Medicine <i>Research Associate</i>	<i>May 2023 - March 2024</i> New York, NY
Eaton Corporation <i>Data Scientist Intern</i>	<i>June 2019 - September 2019</i> Shanghai, China

Honors/Awards

School of Information & Computer Science Chair's Award <i>Donald Bren School of Information and Computer Sciences</i>	Irvine, CA <i>September 2024</i>
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Publications

Understanding Stigmatizing Language Lexicons: A Comparative Analysis in Clinical Contexts Yiliang Zhou, Di Hu, Tianchu Lyu, Jasmine Dhillon, Alexandra L. Beck, Gelareh Sadigh, Kai Zheng. *AMIA Annual Symposium Proceedings*.

EvidenceOutcomes: A Dataset of Clinical Trial Publications with Clinically Meaningful Outcomes Yiliang Zhou*, Abigail M. Newbury*, Gongbo Zhang, Betina Ross Idnay, Hao Liu, Chunhua Weng, Yifan Peng. *MEDINFO 2025, Studies in Health Technology and Informatics*, 329:723–727. *Co-first authors.

Enhancing Disease Detection in Radiology Reports Through Fine-tuning Lightweight LLM on Weak Labels Yishu Wei, Xindi Wang, Hanley Ong, Yiliang Zhou, Adam Flanders, George Shih, Yifan Peng. *AMIA Joint Summits on Translational Science Proceedings*, 2025:614–623.

Evaluating GPT-4V (GPT-4 with Vision) on Detection of Radiologic Findings on Chest Radiographs Yiliang Zhou, Hanley Ong, Patrick Kennedy, Carol C. Wu, Jacob Kazam, Keith Hentel, Adam Flanders, George Shih, Yifan Peng. *Radiology*, 311(2):e233270, 2024.

A Span-Based Model for Extracting Overlapping PICO Entities from Randomized Controlled Trial Publications Gongbo Zhang, Yiliang Zhou, Yan Hu, Hua Xu, Chunhua Weng, Yifan Peng. *Journal of the American Medical Informatics Association (JAMIA)*, 31(5):1163–1171, 2024.

Deciphering Genomic Codes Using Advanced Natural Language Processing Techniques: A Scoping Review Shuyan Cheng, Yishu Wei, Yiliang Zhou, Zihan Xu, Drew N. Wright, Jinze Liu, Yifan Peng. *Journal of the American Medical Informatics Association (JAMIA)*, 32(4):761–772, 2025.

A Natural Language Processing Approach to Detect Inconsistencies in Death Investigation Notes Attributing Suicide Circumstances Song Wang, **Yiliang Zhou**, Ziqiang Han, Cui Tao, Yunyu Xiao, Ying Ding, Joydeep Ghosh, Yifan Peng. *Communications Medicine*, 4:199, 2024.

Closing the Gap Between Open-Source and Commercial Large Language Models for Medical Evidence Summarization Gongbo Zhang, Qiao Jin, **Yiliang Zhou**, Song Wang, Betina Idnay, Yiming Luo, Elizabeth Park, Jordan G. Nestor, Matthew E. Spotnitz, Ali Soroush, Thomas R. Campion Jr., Zhiyong Lu, Chunhua Weng, Yifan Peng. *npj Digital Medicine*, 7:239, 2024.

Towards Long-Tailed, Multi-Label Disease Classification from Chest X-ray: Overview of the CXR-LT Challenge Gregory Holste, **Yiliang Zhou**, Song Wang, Ajay Jaiswal, Mingquan Lin, et al. *Medical Image Analysis*, 97:103224, 2024.

Hidden Flaws Behind Expert-Level Accuracy of Multimodal GPT-4 Vision in Medicine Qiao Jin, Fangyuan Chen, **Yiliang Zhou**, Ziyang Xu, Justin M. Cheung, Robert Chen, Ronald M. Summers, et al. *npj Digital Medicine*, 7:190, 2024.

Generative Large Language Models Trained for Detecting Errors in Radiology Reports Cong Sun*, Kurt Teichman*, **Yiliang Zhou**, Brian Critelli, David Nauheim, Graham Keir, Xindi Wang, Judy Zhong, Adam E. Flanders, George Shih, Yifan Peng. *Radiology*, 315(2):e242575, 2025.

Establishing Dermatopathology Encyclopedia DermPathNet with Artificial Intelligence-Based Workflow Ziyang Xu*, Mingquan Lin*, **Yiliang Zhou**, Zihan Xu, Seth J. Orlow, Shane A. Meehan, Alexandra Flamm, Ata S. Moshiri, Yifan Peng. *Scientific Data*, 13:368, 2026.

A Systematic Review of Early Evidence on Generative AI for Drafting Responses to Patient Messages Di Hu, Yawen Guo, **Yiliang Zhou**, Lidia Flores, et al., Kai Zheng. *npj Health Systems*, 2:27, 2025.

Applying Natural Language Processing and Large Language Models to Clinical Notes for Phenotyping and Diagnosing Rare Diseases: A Systematic Review Seungjun Kim, **Yiliang Zhou**, Yawen Guo, Changrui Xiao, Kai Zheng. *Journal of the American Medical Informatics Association (JAMIA)*, 33(6):1225–1235, 2026.

A Scoping Review of Studies on Secure Messaging through Patient Portals: Persistent Challenges and Potential Solutions Yawen Guo, Di Hu, **Yiliang Zhou**, Tianchu Lyu, Ziqi Yang, Tera L. Reynolds, Kai Zheng. *npj Health Systems*, 2026.

Computational Use of Patient–Provider Secure Messaging Data to Achieve Better Clinical Efficiency and Quality of Communication: A Systematic Review Yawen Guo, Di Hu, **Yiliang Zhou**, Tianchu Lyu, Kai Zheng. *AMIA Annual Symposium Proceedings*, 2025.

From Conversation to Chart: An Analysis of Clinician Edits to Ambient AI Draft Notes Yawen Guo, Di Hu, **Yiliang Zhou**, Tianchu Lyu, Sairam Sutari, Steven Tam, Emilie Chow, Danielle Perret, Deepti Pandita, Kai Zheng. *medRxiv*, 2026.

Understanding Clinician Edits to Ambient AI Draft Notes: A Feasibility Analysis Using Large Language Models Yawen Guo, **Yiliang Zhou**, Di Hu, Sairam Sutari, Emilie Chow, Steven Tam, Danielle Perret, Deepti Pandita, Kai Zheng. Accepted by *AMIA Annual Symposium*. Preprint available on *medRxiv*, 2026.

Enhancing Health Fact-Checking with LLM-Generated Synthetic Data Jingze Zhang, Jiahe Qian **Yiliang Zhou**, Yifan Peng. *arXiv*, 2025.

Understanding Stigmatizing Language in Clinical Documentation: A Paired Comparison of Ambient AI Drafts and Clinician Finalized Notes **Yiliang Zhou**, Yawen Guo, Sairam Sutari, et al., Kai Zheng. Accepted by *AMIA Annual Symposium*. Preprint, 2026.

Examine Clinicians’ Modification of Hedging Language in Ambient AI Documentation: A Comparative Study of AI Drafts and Final Notes **Yiliang Zhou**, Yawen Guo, Di Hu, et al., Kai Zheng. Accepted by *AMIA Annual Symposium*. Preprint, 2026.

Grants

NIH/NIAID Grant 1U19AI181968 (Felgner)

Graduate Student Researcher, Data Management Core
UCI Vaccines for Pandemic Preparedness Center (VPPC)
Total Funding: \$33,116,067

August 2024 – July 2027

Professional Service

Association for Computational Linguistics (ACL)	
Reviewer	2026
AMIA Annual Symposium	
Reviewer	2025 – Present
IEEE ICHI	
Reviewer	2025
npj Digital Medicine	
Ad Hoc Reviewer	2026
Medical Physics	
Ad Hoc Reviewer	2024 – Present

Professional Memberships

American Medical Informatics Association (AMIA)
Member

Technical Skills

Programming Languages/Tools	Java, Javascript, Python, R, Numpy, Pandas, Sklearn, Pytorch, Tensorflow, CUDA
Data Analysis	R studio, MySQL, PostgreSQL